

Synthesis of Long-Range Dependent Symbol Sequences

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1 Annotated Bibliography

This document starts the literature foundation for S5.jl, a package for generating long-range dependent non-numerical symbol sequences. The present version is an annotated bibliography rather than a full paper draft. Manually downloaded PDFs live under `paper/downloads/`; open PDFs fetched during bibliography work live under `paper/references/`. Items not yet downloaded, partial downloads, and alternative matches are tracked in `DOWNLOAD.mn`.

1.1 Long-Range Dependence and Self-Similarity

Samorodnitsky (2006). *Long range dependence*. This survey is a concise bridge between probability theory and applied long-memory models. It is useful for S5.jl because it separates several meanings that are easy to blur in finite simulations: covariance-based long memory, heavy tails, self-similarity, and weak convergence limits. The distinction supports the package policy of documenting whether a generator has an exact, asymptotic, latent, nominal, or empirical LRD claim.

Leland, Taqqu, Willinger, and Wilson (1994). *On the self-similar nature of Ethernet traffic*. This paper is one of the central empirical starting points for modern LRD network-traffic modelling. It showed that packet traffic could retain burstiness across aggregation scales, motivating models beyond Poisson and short-memory Markov assumptions. For S5.jl it supplies the broader methodological lesson: synthetic test data should reproduce large-scale structure, not just local statistics.

Pipiras and Taqqu (2017). *Long-Range Dependence and Self-Similarity*. This monograph provides the mathematical background tying together covariance decay, spectral poles, self-similarity, and the Hurst parameter. It is the main reference for the parameter relationships used throughout the package, including the links between spectral exponent, ACF decay exponent, and Hurst parameter.

Beran (1994). *Statistics for Long-Memory Processes*. Beran's text is a core statistical reference for inference and modelling in long-memory time series. Although S5.jl is not an estimator package, the book helps frame why estimator benchmarking needs synthetic sequences with known parameters and known violations of short-memory assumptions.

Taqqu, Teverovsky, and Willinger (1995). *Estimators for long-range dependence: an empirical study*. This empirical comparison of LRD estimators motivates the separation between S5.jl and a future estimator package. The paper demonstrates that estimator performance depends

heavily on model class, sample size, and diagnostics, which is why S5.jl focuses first on controllable generators.

Witt and Malamud (2013). *Quantification of long-range persistence in geophysical time series: conventional and benchmark-based improvement techniques.* This survey is useful even though its main examples are geophysical rather than symbolic. It emphasizes that finite records, trends, nonstationarities, and method choices can create misleading apparent persistence. That caution maps directly to S5.jl’s validation policy: finite generated sequences should be benchmarked against known controls, not treated as proof of asymptotic LRD.

Malamud and Turcotte (1999). *Self-affine time series: I. generation and analyses.* This review is a compact reference for generating and analyzing self-affine numerical time series. It is relevant to the PB generator family because it collects practical relationships among spectral slopes, Hurst parameters, and finite-sample diagnostics that are often used before symbolic quantization.

1.2 Numerical LRD Synthesis

Paxson (1997). *Fast, approximate synthesis of fractional Gaussian noise for generating self-similar network traffic.* This is the direct ancestor of S5.jl’s PB1 method. It gives a practical spectral construction of fGn, trading exact covariance matching for speed. In S5.jl the numerical fGn sequence is mapped to symbols by rank binning, preserving a simple path from target Hurst parameter to categorical output.

Dieker (2004). *Simulation of fractional Brownian motion.* Dieker’s thesis surveys exact and approximate fBm/fGn simulation methods, including circulant embedding. It provides the natural next reference for improving PB1 if exact covariance control becomes more important than speed.

Coeurjolly (2000). *Simulation and identification of the fractional Brownian motion: a bibliographical and comparative study.* This paper is a methodological survey of fBm simulation and estimation methods. For S5.jl it is a useful checklist for future PB1/PB3 validation, because it compares multiple synthesis pathways rather than treating one fGn construction as canonical.

Wood and Chan (1994). *Simulation of stationary Gaussian processes in $[0, 1]^d$.* Wood and Chan provide a classic circulant-embedding construction for exact stationary Gaussian simulation under appropriate embedding conditions. This is a key reference if S5.jl later adds an exact-covariance Gaussian backend before mapping numerical samples to symbols.

Dietrich and Newsam (1997). *Fast and exact simulation of stationary Gaussian processes through circulant embedding of the covariance matrix.* This work complements Wood and Chan by developing fast exact Gaussian-process simulation through covariance embedding. It is relevant to PB1/PB2 as a possible upgrade path when exact covariance control matters more than the current fast approximate spectral path.

Craigmile (2000, 2003). *Simulating a class of stationary Gaussian processes using the Davies-Harte algorithm, with application to long memory processes.* Craigmile gives a modern treatment of the Davies–Harte method for stationary Gaussian long-memory simulation. The local download is the 2000 NRCSE technical report version; the 2003 Journal of Time Series Analysis article is

the later published version. This is a natural reference for comparing PB1’s approximate fGn construction against an exact or near-exact stationary Gaussian baseline.

Sørbye, Myrvoll-Nilsen, and Rue (2017). *An approximate fractional Gaussian noise model with $O(n)$ computational cost.* This paper approximates fGn with a weighted sum of independent AR(1) components. It matters for S5.jl because it suggests a scalable path between a simple finite-state model and an fGn-like covariance target: a small mixture of short-memory components can approximate long-memory ACFs well over practical finite ranges.

Roughan, Veitch, and Abry (2000). *Real-time estimation of the parameters of long-range dependence.* This work is important for the wavelet perspective behind the planned PB3 generator. Wavelet methods are attractive because they align naturally with multi-scale structure and may provide a bridge from LRD synthesis to local state-machine control. The local download is an extended version of the IEEE/ACM Transactions on Networking paper, with a related earlier GLOBECOM paper also retained as supplemental context.

Bai and Shami (2013). *Modeling self-similar traffic for network simulation.* This paper surveys and evaluates practical self-similar traffic models for network simulation. It is useful implementation context for S5.jl because it keeps the focus on reproducible generators whose outputs are used downstream by other experiments, rather than on estimation alone.

Millán (2021). *On the generation of self-similar with long-range dependent traffic using piecewise affine chaotic one-dimensional maps.* This recent generator paper revisits chaotic-map mechanisms for LRD traffic. It is not directly a symbolic generator, but it is relevant to MB-style mechanisms: simple one-dimensional deterministic or nearly deterministic state updates can produce tunable apparent LRD over finite scale ranges.

1.3 Point Processes and Heavy-Tailed Models

Garrett and Willinger (1994). *Analysis, modeling, and generation of self-similar VBR video traffic.* This paper is a classic example of generating self-similar traffic from heavy-tailed source behaviour. It motivates MB2’s heavy-tailed regime-sojourn construction, where long periods in a regime create large-scale dependence while the within-regime Markov chain controls local symbol structure.

Lowen and Teich (1995, 2005). *Estimation and simulation of fractal stochastic point processes.* Lowen and Teich provide the point-process foundation for fractal renewal and shot-noise constructions. The exact 1995 Fractals article has not yet been downloaded, but a local partial preview of their later book *Fractal-Based Point Processes* is available and covers closely related background. Their work informs MB3’s view of each symbol stream as an independent heavy-tailed arrival process.

Turner, Lowen, Feurstein, Heneghan, Feichtinger, and Teich (1997). *Analysis, synthesis, and estimation of fractal-rate stochastic point processes.* This locally downloaded arXiv paper is not a replacement for the exact Lowen and Teich (1995) article, but it is a closely related simulation and estimation study from the same point-process research line. It is useful context for finite-sample bias in simulated fractal-rate point processes and therefore for S5.jl’s cautious validation stance.

Ryu and Lowen (1998). *Point process models for self-similar network traffic, with applications.* This paper develops point-process models that connect heavy-tailed mechanisms to self-similar traffic. It is a key bridge between network-traffic synthesis and the symbolic FSS construction used in S5.jl.

Roughan, Yates, and Veitch (1999). *The mystery of the missing scales: pitfalls in the use of fractal renewal processes to simulate LRD processes.* This paper is a warning label for naive heavy-tailed simulation. It shows that FRP constructions can fail to express LRD over the intended scale range. S5.jl’s validation scripts and TODO list retain this as an explicit design concern for FSS and On/Off methods.

1.4 Symbolic and Categorical Models

1.4.1 DNA and Genomic Sequence Applications

Peng et al. (1992). *Long-range correlations in nucleotide sequences.* This Nature paper is one of the standard starting points for the DNA LRD literature. It treats nucleotide strings as symbolic sequences whose large-scale fluctuations can differ between coding and noncoding regions, making it a direct application-side precedent for S5.jl’s interest in categorical long memory.

Li and Kaneko (1992). *Long-range correlation and partial $1/f$ spectrum in a noncoding DNA sequence.* This paper is an early noncoding-DNA contribution to the same debate. It is important for S5.jl because it links categorical sequence structure, spectra, and genome organization while also reminding us that a fitted spectral slope is an empirical claim about a finite symbolic record.

Voss (1992). *Evolution of long-range fractal correlations and $1/f$ noise in DNA base sequences.* Voss is an early and especially important symbolic-sequence paper. The paper criticizes low-dimensional DNA walks because arbitrary numerical encodings can introduce artificial correlations, and instead decomposes a symbolic sequence into one binary indicator sequence per base. The resulting equal-symbol correlation and spectral-density summaries are very close in spirit to S5.jl’s centered one-hot validation diagnostics. Voss then applies the method to GenBank categories and reports low-frequency $1/f^\beta$ behavior together with short-range periodicities such as codon-scale peaks. For S5.jl, the main lesson is methodological: symbolic LRD diagnostics should be defined at the symbol level before any numerical estimation is applied.

Buldyrev et al. (1998). *Analysis of DNA sequences using methods of statistical physics.* This review summarizes DNA-walk, fluctuation, and detrended-fluctuation approaches to noncoding and coding DNA. It is useful for S5.jl because it shows how symbolic biological data are commonly transformed before LRD diagnostics, and because it emphasizes finite-length, patchiness, and coding/noncoding distinctions.

Li, Marr, and Kaneko (1994). *Understanding long-range correlations in DNA sequences.* This early review-style paper is directly relevant to non-numerical LRD because DNA is naturally categorical. It connects empirical DNA correlations with expansion-modification mechanisms, making it one of the clearest precedents for symbolic generators whose memory is not merely inherited from a latent numeric time series.

Buldyrev (2006/2013). *Power law correlations in DNA sequences.* This book chapter reviews power-law correlation evidence and interpretation for DNA, with emphasis on noncoding DNA, patchiness, repeated sequences, and duplication-related evolutionary mechanisms. It is valuable as a broad application survey and as a source of further DNA references, but it should be used as review context rather than as a single definitive statistical claim.

Salgado-García and Ugalde (2012). *Exact scaling in the expansion-modification system.* This paper analyzes a binary symbolic mutation-replication model and proves power-law correlation behavior over a range of mutation probabilities. It is especially useful for S5.jl because it gives a mathematically cleaner symbolic generation mechanism than many empirical DNA studies.

Colliva et al. (2014). *Ising model description of long-range correlations in DNA sequences.* This paper models DNA correlations using an Ising-style construction. It is a useful reminder that symbolic biological sequences can be studied through both direct categorical encodings and latent energy/state models.

1.4.2 Text and Language Applications

Schenkel, Zhang, and Zhang (1993). *Long range correlation in human writings.* This early application paper studies human writing directly as a symbolic sequence. It is useful historical context for the later literature because it places text alongside DNA as a domain where long correlations are measured in strings rather than in naturally numerical observations.

Ebeling and Pöschel (1994). *Entropy and long range correlations in literary English.* This paper studies long-range correlations in text using symbolic letters and entropy-like quantities. It supports an S5.jl use case beyond numerical ACFs: controlled symbolic sequences can help test whether entropy-rate, mutual information, and block-complexity diagnostics recover known long-context structure.

Montemurro and Pury (2002). *Long-range fractal correlations in literary corpora.* This paper maps literary corpora to word-level symbolic time series and reports long-range correlations beyond sentence-scale syntax. It is particularly useful for S5.jl because it keeps words, rather than arbitrary letter encodings, as the basic objects being analyzed.

Alvarez-Lacalle et al. (2006). *Hierarchical structures induce long-range dynamic correlations in written texts.* This PNAS paper gives an application-side explanation for text correlations: large-scale document organization can induce persistent correlations at smaller symbolic levels. That mechanism is closely related to why S5.jl keeps hierarchical and regime-based generators separate from simple finite-order Markov structure.

Altmann, Pierrehumbert, and Motter (2009). *Beyond word frequency: bursts, lulls, and scaling in the temporal distributions of words.* This paper studies recurrence-time structure for words, emphasizing bursty word use and scaling that cannot be inferred from Zipf frequencies alone. It is an application-focused bridge between natural-language data and the bursty symbolic generators in S5.jl.

Altmann, Cristadoro, and Degli Esposti (2012). *On the origin of long-range correlations in texts.* This paper argues that correlations in text can flow from high-level semantic or topic organization down to words and letters. It is a useful companion to the Hawkes-style and hierarchical models because it frames word occurrences as bursty events tied to semantically meaningful structure.

Drożdż et al. (2016). *Quantifying origin and character of long-range correlations in narrative texts.* This paper studies sentence-length variability and reports spectral and multifractal evidence for long-range structure in narrative texts. It is a valuable application reference, especially when paired with Zeng et al. (2025), because it shows both the attraction and the risk of interpreting finite text statistics as LRD evidence.

Tanaka-Ishii and Bunde (2016). *Long-range memory in literary texts: on the universal clustering of the rare words.* This paper focuses on rare-word clustering in literary texts. It is relevant to S5.jl because rare-symbol recurrence can be a stronger application diagnostic than aggregate autocorrelation after an arbitrary numeric encoding.

Ogura, Hanada, Amano, and Kondo (2022). *Modeling long-range dynamic correlations of words in written texts with Hawkes processes.* This paper models word occurrence signals in written texts as Hawkes processes, separating words with observable dynamic correlations from words that behave closer to uncorrelated background processes. It is directly relevant to S5.jl’s symbolic focus because it treats word appearances as binary categorical time-series events and validates simulated point-process outputs against word occurrence autocorrelation functions. The MB4 ‘HawkesSymbol’ generator is a finite-history discrete-time adaptation of this self-exciting point-process idea, not a fitted text model.

Montemurro and Degli Esposti (2026). *A Zipf-preserving, long-range correlated surrogate for written language and other symbolic sequences.* This recent paper is highly aligned with S5.jl’s symbolic goals: it constructs surrogate sequences that preserve empirical symbol frequencies while imposing long-range scaling through an fGn-based assignment. It is both a recent text/DNA application reference and a reminder that preserving marginals and preserving correlation structure are separate controls.

1.4.3 Symbolic Generative and Analytical Models

Li (1989, 1991) and Li (1992). *Expansion-modification systems and replication-mutation generators.* Li’s expansion-modification line is one of the most natural symbolic simulation families for S5.jl to revisit. The mechanism repeatedly copies and mutates symbols or blocks, producing symbolic sequences with spatial $1/f$ spectra and long correlations. It is especially attractive for DNA-like applications because the mechanism is symbolic from the start and has a biological interpretation via duplication and mutation.

Melnyk, Usatenko, and Yampol’skii (2006). *Memory functions of the additive Markov chains: applications to complex dynamic systems.* This paper gives an explicit route from a desired pair-correlation function to a memory function for additive Markov chains. It is an important candidate model for S5.jl because it suggests a more principled future version of LAMP/MB1: instead of choosing a power-law history kernel directly, one can derive or approximate a memory function for a target symbolic correlation profile.

Mayzelis et al. (2006). *Additive N -step Markov chains as prototype model of symbolic stochastic dynamical systems with long-range correlations.* This paper generalizes binary additive Markov chains, including biased symbol frequencies, and verifies analytical correlation and word-count variance results by simulation. It supports an S5.jl future path toward additive Markov generators with clearer correlation-function control than the current finite-history LAMP interface.

Melnik and Usatenko (2014). *Entropy and long-range correlations in random symbolic sequences.* This paper develops entropy estimates for finite-alphabet long-correlated symbolic sequences and applies them to written English and DNA nucleotide sequences. It is useful for S5.jl because it connects high-order additive Markov models, pair correlations, and entropy diagnostics in the same symbolic setting.

Gal, Chen, and Ghahramani (2015). *Latent Gaussian processes for distribution estimation of multivariate categorical data.* This work motivates PB2. The central idea is to represent categorical outcomes through latent Gaussian variables and an argmax transformation. S5.jl adapts that idea to long-memory latent fGn streams and adds finite-sample offset calibration for marginal control.

Kumar, Raghu, Sarlos, and Tomkins (2017). *Linear additive Markov processes.* This paper motivates MB1. LAMP models make transition probabilities additive in past observations, allowing long-range memory to be encoded through a decay kernel over history. S5.jl implements a finite-depth version with a marginal innovation term. The correct WWW 2017 citation uses DOI 10.1145/3038912.3052644; earlier notes with a different suffix were stale.

Koroteev (2015). *On a chain of fragmentation equations for duplication-mutation dynamics in DNA sequences.* This paper studies duplication-mutation dynamics and exact-repeat length distributions in synthetic and natural DNA. It is not directly an LRD generator paper, but it strengthens the case for a future DNA-oriented symbolic generator based on block duplication, mutation, and finite-sequence provenance.

Provata and Beck (2012). *Coupled intermittent maps modelling the statistics of genomic sequences: a network approach.* This paper uses coupled intermittent maps to generate symbolic sequences with DNA-like long-range features. It is an interesting candidate mechanism, but it is less immediately aligned with S5.jl than additive Markov or duplication-mutation models because the controllable user-facing parameters would need careful translation from map dynamics to alphabet, marginal, and local-structure contracts.

Singh, Greenberg, and Klakow (2016). *The custom decay language model for long range dependencies.* This language-modelling paper is closely related to LAMP in spirit, but its mechanism is more semantic and predictive than S5.jl’s present generators. CDLM combines a short-distance N -gram language model with a log-linear long-distance component over word vectors, transformed by distance-specific matrices T_k . Those matrices allow a custom decay of contextual influence rather than a single scalar decay. S5.jl does not implement a word-vector language model, but the paper supports the broader MB1 design idea: finite symbolic histories can be weighted by explicit distance kernels while keeping the parameter growth far below that of high-order Markov models.

Dębowski (2015). *Hilberg exponents: new measures of long memory in the process.* This paper develops power-law growth rates for mutual information and universal codes. It is relevant to S5.jl because symbolic LRD need not be visible only as an autocorrelation of arbitrary numeric encodings; information growth can be a more natural diagnostic for text-like and categorical processes.

Carpio and Daley (2007). *Long-range dependence of Markov chains in discrete time on countable state space.* This paper is important because it frames LRD for Markov chains through counting and recurrence-time behaviour. It helps justify the use of count-process and return-time definitions for non-numerical sequences where ordinary numerical ACF methods are not directly applicable.

Zeng, Cui, and Li (2025). *Fractal illusions: an experimental study of long-range sentence-length correlations in randomly generated natural language texts.* This recent paper is valuable as a cautionary reference. It argues that apparent sentence-length long-range correlations can arise in randomized text generation, so symbolic LRD diagnostics need controls that distinguish genuine memory mechanisms from artifacts of preprocessing, punctuation, and measurement.

Dębowski (2025). *From Zipf’s law to neural scaling through Heaps’ law and Hilberg’s hypothesis.* This recent short paper connects power laws in symbolic language data to large-scale neural language-model behavior. For S5.jl it is motivation for keeping symbolic, information-theoretic, and machine-learning validation paths visible even though the package itself remains a generator library.

Wieczyński and Dębowski (2025). *Long-range dependence in word time series: the cosine correlation of embeddings.* This recent paper studies word sequences through embedding-based correlations rather than direct numeric observations. It defines LRD through slower than exponential decay of two-point Shannon mutual information and uses a cosine correlation of word2vec embeddings as a tractable lower-bound proxy. The authors report a visible stretched-exponential decay for natural texts up to lags around 1000 words, while a human-vs-LLM corpus does not show the same systematic signal. This suggests a possible validation direction for symbolic generators whose alphabet elements later receive semantic or learned representations, but it is not yet part of S5.jl’s core validation path.

2 Model Details and S5.jl Adaptations

This section records the concrete model ideas suggested by the downloaded literature and how S5.jl adapts them. The distinction is important: many papers study diagnostics or fitted models for real data, whereas S5.jl implements reproducible generators with explicit finite-sample controls.

2.1 Indicator and Spectral Symbol Diagnostics

Voss (1992) is not a generator paper, but it is central to how symbolic LRD should be measured. The paper defines one indicator sequence $U_k[x_n]$ for each symbol k and computes equal-symbol correlations and spectra from those indicator sequences. This avoids assigning arbitrary numeric values to bases. S5.jl modifies this diagnostic idea by using centered one-hot indicator series in validation:

$$x_t^{(k)} = 1\{X_t = k\} - \bar{p}_k.$$

Centering removes the symbol marginal from the zero-frequency component before autocorrelation or periodogram calculations. The implementation also averages over symbols and replicates, drops undefined constant indicators, and marks finite-sample and generator-memory interpretation limits on plots. Thus Voss’s equal-symbol analysis becomes a reproducible validation transformation rather than a generator API.

2.2 PB1: Spectral fGn Plus Quantization

Paxson (1997), together with the fGn simulation surveys by Dieker (2004), Coeurjolly (2000), Wood and Chan (1994), Dietrich and Newsam (1997), and Craigmile (2000/2003), supplies the numerical simulation background for PB1. The literature model is a real-valued Gaussian process with a target long-memory spectrum or covariance. S5.jl modifies this into a symbolic generator by synthesizing approximate fGn and then rank-binning the sample into alphabet symbols. The rank bins are chosen from requested marginal probabilities, giving finite-sample counts as close as possible to the target marginal. The modification is deliberately labeled property-based: the LRD structure lives in a latent numerical layer and is inherited only after quantization.

2.3 PB2: Latent Gaussian Categorical Mapping

Gal, Chen, and Ghahramani (2015) model multivariate categorical observations by placing latent Gaussian structure behind a softmax-like categorical observation model. Their paper is about Bayesian distribution estimation for categorical vectors, not LRD time-series synthesis. S5.jl adapts the idea by using one latent fGn stream per symbol and choosing the emitted symbol by an argmax over the latent values plus calibrated offsets. The calibration offsets target the requested marginal distribution in finite samples. This keeps the useful latent-Gaussian categorical mechanism while replacing the original inference problem with a reproducible sequence generator.

2.4 PB3: Multiscale Driver with Markov Regimes

The wavelet and multiscale LRD literature motivates PB3, but S5.jl treats the current implementation as a practical generator rather than a completed wavelet-calibrated theory. The implemented model draws a latent long-memory driver, bins that driver into regimes, and lets each regime select a Markov transition matrix for the visible symbols. The original Haar-like cascade is retained as a comparison path, while the default uses a spectral fGn driver because validation showed that the earlier cascade produced overly flat autocorrelation. The S5.jl modification is therefore twofold: it makes the observable local structure explicit through Markov matrices, and it keeps the driver choice visible in provenance.

2.5 PB4: Intermittent Map Symbols

Provata and Beck (2012) use intermittent polynomial maps, including coupled map lattices, to model genomic symbolic statistics. Their construction maps a continuous intermittent dynamical system into DNA symbols by partitioning phase space so that symbol frequencies match the target sequence. S5.jl implements a one-dimensional intermittent-map generator, then rank-bins the latent trajectory into arbitrary alphabets with requested marginal probabilities. It does not yet implement the coupled-map network machinery or fit chromosome-specific partition boundaries. This keeps PB4 a simple property-based generator with a latent dynamical source of long laminar episodes.

2.6 MB1a and MB1b: LAMP-Style Additive Histories

Kumar, Raghu, Sarlos, and Tomkins (2017) define LAMP by selecting a lag from a history distribution w and then applying a transition matrix P from the state at that lag. Their main aim is a learnable sequence model with long context and relatively few parameters compared with high-order Markov chains. S5.jl adapts this into MB1a by using an explicit finite history depth d , a power-law lag distribution, a user-visible transition matrix, and an innovation term toward the declared marginal distribution. The innovation term is an S5.jl engineering modification to avoid finite-history absorption and keep marginal control visible.

MB1b is a scalability modification rather than a separate literature model. It groups old lags into dyadic buckets and uses bucket-level symbol mixtures instead of scanning every lag. The scientific identity remains LAMP-like additive history weighting, but the implementation trades exact finite-history weights for a scalable approximation whose cutoff and approximation status are documented.

2.7 MB1c: Additive Markov Memory Functions

Mayzelis et al. (2006) and Melnyk, Usatenko, and Yampol’skii (2006) develop additive N -step Markov chains where conditional symbol probabilities are linear functions of previous symbols, and where memory functions can be related to desired correlation functions. S5.jl’s MB1c is the closest implemented path to this literature. It uses a centered additive memory function around the target marginal:

$$q(s) = p(s) + \rho \sum_{r=1}^d w_r (1\{X_{t-r} = s\} - p(s)),$$

with finite cutoff d , normalized power-law weights, and a strength parameter ρ . The current implementation does not solve the inverse problem of recovering a memory function from a target correlation curve; that remains a future calibration path.

2.8 CDLM as a Future Text-Oriented Additive Model

Singh, Greenberg, and Klakow (2016) propose CDLM, a language model combining a short-distance trigram component with a log-linear long-distance term over word vectors and distance-specific transition matrices T_k . S5.jl does not implement CDLM because it would require a vocabulary embedding model and a predictive language-modelling objective, both outside the current generator scope. The useful idea for S5.jl is the separation of short-range symbolic detail from a parametric long-range decay mechanism. A future text-oriented generator could adapt this by replacing word embeddings with user-supplied symbol features while preserving S5.jl’s explicit RNG, provenance, and control contracts.

2.9 MB2 and MB3: Heavy-Tailed Regimes and Renewal Streams

Garrett and Willinger (1994), Lowen and Teich’s point-process work, Ryu and Lowen (1998), and Roughan, Yates, and Veitch (1999) motivate heavy-tailed source and renewal mechanisms. In the literature, LRD arises because on/off periods or renewal intervals have heavy-tailed distributions, and superposition or aggregation can produce self-similar traffic. S5.jl adapts this in two symbolic directions. MB2 uses heavy-tailed regime sojourns while emissions within each regime follow explicit Markov transition matrices. MB3 assigns each symbol its own renewal stream and emits the symbol with the next pending event. S5.jl keeps the Roughan–Yates–Veitch warning visible: finite samples

can miss the largest renewal scales, so validation plots mark interpretation limits and the package does not treat a finite simulation as proof of LRD.

2.10 MB4: Hawkes-Style Symbol Excitation

Ogura et al. (2022) model word occurrence signals as Hawkes processes, with Type-I words showing bursty long-range dynamic correlations and Type-II words behaving closer to uncorrelated background. Their model is fitted to binary occurrence signals for individual words. S5.jl modifies the idea into a discrete-time finite-history symbolic generator: every alphabet symbol has a baseline intensity, each recent symbol contributes a power-law weighted row of an excitation matrix, and one symbol is sampled at each time step. This is a finite-history symbolic analogue, not a fitted continuous-time Hawkes process.

2.11 MB5: Duplication-Mutation Growth

Li’s expansion-modification work, Salgado-Garcia and Ugalde (2012), and Koroteev (2015) motivate symbolic growth models based on copying and mutation. Koroteev’s downloaded paper focuses on duplication-mutation dynamics in DNA and the length distributions of exact repeats, rather than directly on autocorrelation. S5.jl’s MB5 adapts the broad copy/mutate mechanism to sequence synthesis by selecting a copy lag from a truncated power-law distance kernel and mutating copied symbols with a configured probability. An earlier block-source interpretation was dropped after validation showed nearly flat autocorrelation. The current implementation is therefore a pragmatic power-law-lag copy generator, not a full chromosome-evolution model with insertions, block rearrangements, or exact-repeat length calibration.

2.12 Machine-Learning Benchmarks and Diagnostics

Belletti et al. (2018), Vaswani et al. (2017), Tay et al. (2021), Debowski’s Hilberg-exponent work, and Wieczynski and Debowski (2025) do not directly define S5.jl generators. They motivate why controlled symbolic LRD sequences are useful. Belletti et al. argue that common gated recurrent architectures may fail to sustain LRD-like memory. Long Range Arena standardizes long-context model comparisons, but does not focus on statistically controlled LRD generators. Wieczynski and Debowski use embedding cosine correlation as a proxy for mutual information in word time series. These papers suggest future validation and learning tasks: feed S5.jl sequences to learners, compare local and long-context performance, and later test embedding-based diagnostics for large symbolic alphabets.

2.13 Entropy, Predictability, and Information Storage

Crutchfield and Feldman (2003). *Regularities unseen, randomness observed: Levels of entropy convergence.* This paper is a core reference for excess entropy and entropy-rate convergence. It frames excess entropy as stored predictive information in a process, making it a natural quantity to test on controlled LRD symbolic sequences. S5.jl can create cases where local Markov structure and long-range structure are varied separately, which is exactly the kind of setting needed to probe entropy-rate convergence.

Bialek, Nemenman, and Tishby (2001). *Predictability, complexity, and learning.* This work connects prediction, complexity, and information in sequences. It is useful background for studying

how much long-range structure a learner can extract from finite data, and for interpreting excess entropy-like quantities as predictive information.

2.14 Machine Learning Motivation

Belletti, Beutel, Jain, and Chi (2018). *Factorized recurrent neural architectures for longer range dependence.* This paper motivates the project from the machine-learning side by showing that common recurrent architectures struggle with long-range dependence. Its theoretical argument connects gated recurrent dynamics and bounded nonlinear state updates to short memory under assumptions related to vanishing gradients. The proposed factorized recurrent architecture is not a source model for S5.jl, but it helps explain why synthetic NN-LRD data is useful for stress-testing learning algorithms: a learner can appear strong on local sequence modelling while still failing on dependence that persists over large lags.

Belletti, Chen, and Chi (2019). *Quantifying long range dependence in language and user behavior to improve RNNs.* This paper is one of the direct motivations for studying LRD in language and human behavioural sequences. It connects empirical LRD measurement to model design, reinforcing S5.jl’s role as a controllable data source for later estimator and learning experiments.

Vaswani et al. (2017). *Attention is all you need.* The transformer architecture made long-context neural sequence modelling central to modern machine learning. Although S5.jl is not language-specific, its symbolic generators can create non-language sequences with controlled long-context dependencies for training and stress-testing transformer-style models.

Tay et al. (2021). *Long Range Arena: A benchmark for efficient transformers.* Long Range Arena motivates synthetic and semi-synthetic tasks designed to test long-context modelling. The downloaded preprint describes tasks with inputs roughly from 1K to 16K tokens, including byte-level text, retrieval, image-like data, and synthetic hierarchical reasoning. S5.jl extends this spirit in a more statistical direction: rather than benchmarking transformer variants directly, it produces categorical processes where alphabet, marginal distribution, local Markov structure, and nominal LRD mechanism can be controlled and recorded.

2.15 Two-Dimensional and Spatial Self-Similar Processes

Gelbaum (2012). *Fractional Brownian fields over manifolds.* This work generalizes fractional Brownian fields beyond one-dimensional time. It is outside S5.jl’s current scope, but it is a useful reference for future thinking about spatial categorical fields or image-like symbolic processes where LRD appears across two or more dimensions.

Gelbaum and Titus (2013). *Simulation of fractional Brownian surfaces via spectral synthesis on manifolds.* This short paper provides a spectral-synthesis construction for random fractal surfaces. It belongs in a 2D reference section because it shows how the spectral ideas behind PB1 can be generalized to surfaces while preserving an explicit simulation pathway.

Balcerek et al. (2025). *Two-dimensional fractional Brownian motion: analysis in time and frequency domains.* This recent paper studies two-dimensional fBm with dependent components, including covariance and spectral structure. It is not a target for S5.jl’s current 1D symbolic

API, but it is useful context if the project later considers multi-dimensional symbolic processes or vector-valued latent drivers.

Stein (2002). *Fast and exact simulation of fractional Brownian surfaces.* Stein’s paper is a classic reference for exact simulation of fractional Brownian surfaces. It is a useful benchmark reference for any future extension from one-dimensional symbolic sequences to spatial or image-like symbolic fields.